DATATUNE 2026

Iceberg Ahead:

Navigating the Future of Data Lakes Without Sinking

A beginner-friendly deep dive into Apache Iceberg

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THE PROBLEM

What Goes Wrong in Traditional Data Lakes?

✕ No Reliable Updates

Changing a single record means rewriting entire files — slow and error-prone

THE PROBLEM

What Goes Wrong in Traditional Data Lakes?

🐌 Slow Query Performance

Engines scan everything — even when you only need a tiny slice of data

THE PROBLEM

What Goes Wrong in Traditional Data Lakes?

No Transaction Safety

If a job fails halfway, you're left with corrupted, partial data

THE PROBLEM

What Goes Wrong in Traditional Data Lakes?

🔒 Schema Rigidity

Adding or changing columns often breaks pipelines or requires full rewrites

WHAT IS APACHE ICEBERG?

A Smart Blueprint for Your Data

Apache Iceberg is an open table format — essentially a smart organizational layer that sits on top of your data files in a data lake.

Think of it like the index of a massive library. Without it, finding a book means searching every shelf. With Iceberg, you go straight to the right spot — every time.

Open

Table Format

Works with any storage system (S3, HDFS, GCS)

🎯 **Iceberg brings database-level reliability to the flexibility of data lakes**

HOW IT WORKS

Iceberg's Three-Layer Architecture

01

Catalog Layer

The Table of Contents

Tracks the current state of every table. It's the entry point — tells engines where to find the latest snapshot of your data.

02

Metadata Layer

The Blueprint

Stores snapshots, schema history, and partition info. Each change creates a new snapshot — enabling time travel and rollbacks.

03

Data Layer

The Actual Files

Your real data stored as Parquet, ORC, or Avro files. Iceberg manages these efficiently — only touching what's needed.

TIME TRAVEL

QUERIES

See your data as it was at any point in time

KEY FEATURE #1

Query Data From The Past

Every change in Iceberg creates a new snapshot — like a save point in a video game. You can travel back to ANY of them.

```
-- See sales data from last week
SELECT * FROM sales
FOR SYSTEM_TIME AS OF
  '2025-01-01 09:00:00'
```

- 🔍 Debug data issues after the fact
- 📊 Audit compliance at a specific date
- 🔄 Rollback accidental deletes instantly

KEY FEATURE #2

Schema Evolution — Let Your Data Grow Up

Business needs change. Your data structure should be able to change too — without breaking anything.

Add Column

Add 'phone_number' to orders table — old data gets null, nothing breaks

Rename Column

Rename 'qty' to 'quantity' — queries using old name still work

Reorder Fields

Move columns around without rewriting a single data file

Drop Column

Remove columns safely — historical queries still reference old snapshots

✅ **All schema changes are backwards compatible — zero downtime, zero rewrites**

KEY FEATURES #3 & #4

Reliability + Performance

ACID Transactions

Like a bank transfer — either the whole operation succeeds, or nothing changes. No more half-written files.

- ✓ Atomic — all or nothing
- ✓ Consistent — data stays valid
- ✓ Isolated — concurrent writes are safe
- ✓ Durable — committed = saved

Hidden Partitioning

Iceberg automatically organizes data into smaller chunks (partitions) so queries only scan what they need.

Think of it like a well-organized filing cabinet. Without partitions, you'd flip through every single folder. With Iceberg's smart partitioning, you go straight to the right drawer.

WHY THE BUZZ?

Why is Everyone Jumping Aboard?

Enterprise Ready

Used by Netflix, Apple, LinkedIn at petabyte scale

Cost Efficient

Skip scanning saves compute and storage costs significantly

FASTEST GROWING

Tool Agnostic

Works with Spark, Flink, Hive, Trino, Dremio and more

Open Standard

Vendor-neutral — no lock-in, backed by Apache Foundation

Adopted by:

Netflix | Apple | LinkedIn | Airbnb | Adobe | AWS | Snowflake | Databricks

...and hundreds more companies worldwide, including many Fortune 500 organizations

COMPARISON

Iceberg vs Traditional Formats

Feature	CSV / Parquet (Traditional)	Apache Hive	Apache Iceberg ✓
ACID Transactions	❌ No	⚠️ Limited	✅ Full Support
Time Travel	❌ No	❌ No	✅ Built-in
Schema Evolution	❌ Manual	⚠️ Partial	✅ Seamless
Hidden Partitioning	❌ Manual	⚠️ Basic	✅ Automatic
Concurrent Writes	❌ No	⚠️ Slow locks	✅ Optimistic
Query Speed (large data)	🐌 Slow scan	⚠️ Moderate	✅ Very Fast

COMPARISON

Iceberg vs Delta Lake

Feature	Delta Lake	Apache Iceberg ✓
MetaData Architecture	⚠️ Append Only transaction log	✅ Hierarchical tree
Engine Compatibility	⚠️ Spark First	✅ Engine Agnostic
Schema Evolution	⚠️ Limited	✅ Full & flexible
Partition Evolution	⚠️ No	✅ Yes
File Formats	⚠️ Primarily Parquet	✅ Parquet, ORC, Avro
Catalog & Governance	⚠️ Unity or Hive Metastore	✅ REST catalog

Choose **Apache Iceberg**

- maximum openness
- engine flexibility (Trino + Spark + Snowflake in the same lake)
- Partition/schema evolution without headaches
- Future-proofing against vendor lock-in.

Choose **Delta Lake**

- Your world is heavily Spark- or Databricks-centric
- Value the tightest integration and operational simplicity there
- You're already invested in Unity Catalog and don't need broad multi-engine writes.

In practice today, many teams use **both** via UniForm (Delta readable as Iceberg) or hybrid setups — the formats are converging in features, but architectural philosophy still drives the choice.

GETTING STARTED

Your First Steps With Apache Iceberg

1 Try It In a Sandbox

Use Apache Spark with Iceberg locally or in Google Colab. No infrastructure required — perfect for learning.

```
pip install pyspark
pip install pyiceberg
```

2 Use a Managed Service

AWS Glue, Snowflake, Databricks and Azure all support Iceberg natively — zero setup headaches.

3 Start Small

Convert one non-critical table to Iceberg format. Observe the benefits before expanding.

4 Read the Docs

Iceberg's official documentation is beginner-friendly. The community on Slack is also very welcoming.

WHOs Integrating?

- **AWS** — Comprehensive support: Amazon Athena, EMR (Spark/Flink/Hive/Trino), Glue, S3 Tables (fully managed Iceberg), Redshift Spectrum.
- **Google Cloud** — BigQuery, BigLake native Iceberg support.
- **Snowflake** — Native Iceberg tables; Polaris (open catalog project); full interoperability.
- **Databricks** — Full support (via Tabular acquisition); Unity Catalog + Iceberg.

WHOs Integrating?

- **Dremio** — Deep integration; Arctic catalog; major contributor.
- **Cloudera** — Native in public cloud/on-prem Lakehouse; majority of customers testing/running Iceberg.
- **Starburst** (Trino-based) — Strong support.
- **Confluent** — Tableflow uses Iceberg for Kafka-to-lake streaming.
- **Redpanda** — Iceberg topics for real-time Kafka-to-Iceberg.
- Others: Presto/Trino ecosystems, Flink, Spark (core), Hive (via connectors), Polaris (Snowflake-led open catalog), Nessie, Tabular (now part of Databricks), and more.

WHOs using?

- Netflix — Original creators; core to their data platform for solving big data reliability issues.
- Apple — Ubiquitous across divisions; powers lakehouse architecture at massive scale (hundreds of teams, petabyte tables, ETL to streaming); active contributor.
- Airbnb — Migrated legacy HDFS to Iceberg on S3 for better reliability and performance.
- Adobe — Early adopter (since ~2020); uses in Adobe Experience Platform for billions of events and personalization.
- LinkedIn — Adopted for improved data reliability, query performance, and analytics.

WHOs using?

- Stripe — Embraced for parts of their data platform.
- Expedia — Uses in data platforms.
- Pinterest — Scaling Iceberg adoption (started ~2022); handles massive usage with tools like Gravitino.
- Tencent — Active user and contributor.
- LY Corporation (Japan) — Integrating into on-premises platform at hundreds of petabytes.
- Capital One — Implements lakehouse architecture with Iceberg tables on Snowflake.
- WHOOP — Reimagines data architecture with Snowflake + Iceberg.

KEY TAKEAWAYS

What You Should Remember

01 Iceberg is an open table format that adds reliability, performance, and flexibility to data lakes

02 Time travel, schema evolution, ACID transactions, and smart partitioning are its superpowers

03 It works with the tools you already use — Spark, Hive, Flink, Trino, and cloud platforms

04 Industry adoption is explosive — this is becoming the standard for modern data architecture

🧊 ***Apache Iceberg: Not just a trend — the future of data lakes***

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Thank You!

Questions? Let's navigate data lakes together.

iceberg.apache.org • Join #iceberg on the Apache Slack